WORKSHEET

# Chapter 8: Stem-and-leaf plots

Students sit a geometry exam and obtain the following results (in %):

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 73 | 65 | 91 | 84 | 56 | 84 | 72 | 66 | 48 | 89 | 67 | 50 | 84 |
| 75 | 62 | 30 | 58 | 96 | 98 | 74 | 43 | 79 | 62 | 55 | 84 | 47 |
| 68 | 77 | 85 | 81 | | | | | | | | | |

**1** Draw a stem-and-leaf plot to represent the data.

**2** How many students achieved 50% or higher?

**3** How many students received more than 80%?

**4** How many students scored between 60% and 70%?

**5** How many students scored less than 40%?

The hourly wages of a group of teenagers working part time are recorded below (in $):

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 11 | 25 | 9 | 17 | 13 | 10 | 16 | 8 | 12 | 14 | 7 | 19 |
| 18 | 22 | 8 | 19 | 31 | 13 | 11 | 39 | 13 | 21 | 7 | 9 | 27 |
| 16 | 16 | 21 | 9 | | | | | | | | | |

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**6** Draw a stem-and-leaf plot to represent the data.

**7** How many people are paid less than $10 per hour?

**8** How many people are paid more than $30 per hour?

**9** How many people are paid between $20 and $30 per hour?

Heights of students in a cross-country running team were measured and recorded (in cm):

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 157 | 169 | 162 | 165 | 174 | 159 | 178 | 166 | 154 | 182 | 168 | 158 | 179 |
| 154 | 159 | 167 | 187 | 167 | 178 | 163 | 149 | 173 | 180 | 156 | 168 | 173 |
| 152 | 162 | 169 | 175 | | | | | | | | | |

**10** Draw a stem-and-leaf plot to represent the data.

**11** How many students were less than 160 cm tall?

**12** How many students were at least 180 cm tall?

**13** How many students were between 160 cm and 170 cm tall?

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## Answers

**1**

| Stem | Leaf |
|---|---|
| 3 | 0 |
| 4 | 3 7 8 |
| 5 | 0 5 6 8 |
| 6 | 2 2 5 6 7 8 |
| 7 | 2 3 4 5 7 9 |
| 8 | 1 4 4 4 4 5 9 |
| 9 | 1 6 8 |

Key: 3|0 = 30%

**2** 26

**3** 10

**4** 6

**5** 1

**6**

| Stem | Leaf |
|---|---|
| 0 | 6 7 7 8 8 9 9 9 |
| 1 | 0 1 1 2 3 3 3 4 6 6 6 7 8 9 9 |
| 2 | 1 1 2 5 7 |
| 3 | 1 9 |

Key: 0|6 = $6

**7** 8

**8** 2

**9** 5

**10**

| Stem | Leaf |
|---|---|
| 14 | 9 |
| 15 | 2 4 4 6 7 8 9 9 |
| 16 | 2 2 3 5 6 7 7 8 8 9 9 |
| 17 | 3 3 4 5 8 8 9 |
| 18 | 0 2 7 |

Key: 14|9 = 149 cm

**11** 9

**12** 3

**13** 11

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